

AP3

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Overview

This project analyzes tangent lines (1D) and tangent planes (2D) for smooth functions using both **exact derivatives** and **finite difference approximations**. Normal vectors are computed from the gradient, and the accuracy of finite difference formulas is evaluated. The study includes tables and plots to illustrate how step size affects error and to compare 1D vs 2D behavior.

Theory

1D Case: Let $f(x)$ be a smooth function.

- Tangent line at x_0 : $y - f(x_0) = f'(x_0)(x - x_0)$
- Tangent direction vector: $\mathbf{t} = [1, f'(x_0)]$
- Normal vector: $\mathbf{n} = [-f'(x_0), 1]$, normalized: $\mathbf{n}/\|\mathbf{n}\|$

2D Case: Let $g(x, y)$ be a smooth function.

- Tangent plane at (x_0, y_0) : $z - g(x_0, y_0) = g_x(x_0, y_0)(x - x_0) + g_y(x_0, y_0)(y - y_0)$
- Tangent directions: $\mathbf{t}_x = [1, 0, g_x]$, $\mathbf{t}_y = [0, 1, g_y]$
- Normal vector: $\mathbf{n} = [g_x, g_y, -1]$, normalized: $\mathbf{n}/\|\mathbf{n}\|$

Finite Difference Approximation: Central difference formula:

$$f'(x) \approx \frac{f(x+h) - f(x-h)}{2h}$$

$$g_x \approx \frac{g(x+h, y) - g(x-h, y)}{2h} \quad g_y \approx \frac{g(x, y+h) - g(x, y-h)}{2h}$$

where h is the step size. Errors are measured by the norm difference between exact and approximate normal vectors.

Code Implementation

- Implement functions $f(x)$, f' exact, $g(x, y)$, g_x , g_y exact.
- Implement central finite difference formulas for 1D derivative and 2D gradient.
- Compute tangent directions and normal vectors using both exact derivatives and finite differences.
- Normalize all vectors for consistent comparison.

Experiments & Results

Step sizes h were varied logarithmically from 10^{-8} to 10^{-1} . Normal and tangent vectors were computed for both 1D and 2D cases, and the **error norm** was recorded:

Step size h	1D Normal Error	2D Normal Error
1.000×10^{-8}	1.985×10^{-10}	1.385×10^{-9}
5.995×10^{-8}	1.142×10^{-10}	1.542×10^{-10}
3.594×10^{-7}	2.695×10^{-11}	1.317×10^{-10}
2.154×10^{-6}	6.363×10^{-13}	1.257×10^{-11}
1.292×10^{-5}	1.875×10^{-11}	1.881×10^{-10}
7.743×10^{-5}	7.091×10^{-10}	6.666×10^{-9}
4.642×10^{-4}	2.549×10^{-8}	2.396×10^{-7}
2.783×10^{-3}	9.161×10^{-7}	8.610×10^{-6}
1.668×10^{-2}	3.294×10^{-5}	3.094×10^{-4}
1.000×10^{-1}	1.205×10^{-3}	1.111×10^{-2}

Step size h	1D Tangent Error	2D Tangent Error
10^{-8}	1.19×10^{-10}	5.10×10^{-10}
5.99×10^{-8}	6.83×10^{-11}	4.80×10^{-11}
3.59×10^{-7}	1.61×10^{-11}	4.28×10^{-11}
2.15×10^{-6}	3.81×10^{-13}	6.37×10^{-12}
1.29×10^{-5}	1.12×10^{-11}	8.66×10^{-11}
7.74×10^{-5}	4.24×10^{-10}	3.08×10^{-9}
4.64×10^{-4}	1.52×10^{-8}	1.11×10^{-7}
2.78×10^{-3}	5.48×10^{-7}	3.98×10^{-6}
1.67×10^{-2}	1.97×10^{-5}	1.43×10^{-4}
10^{-1}	7.20×10^{-4}	5.13×10^{-3}

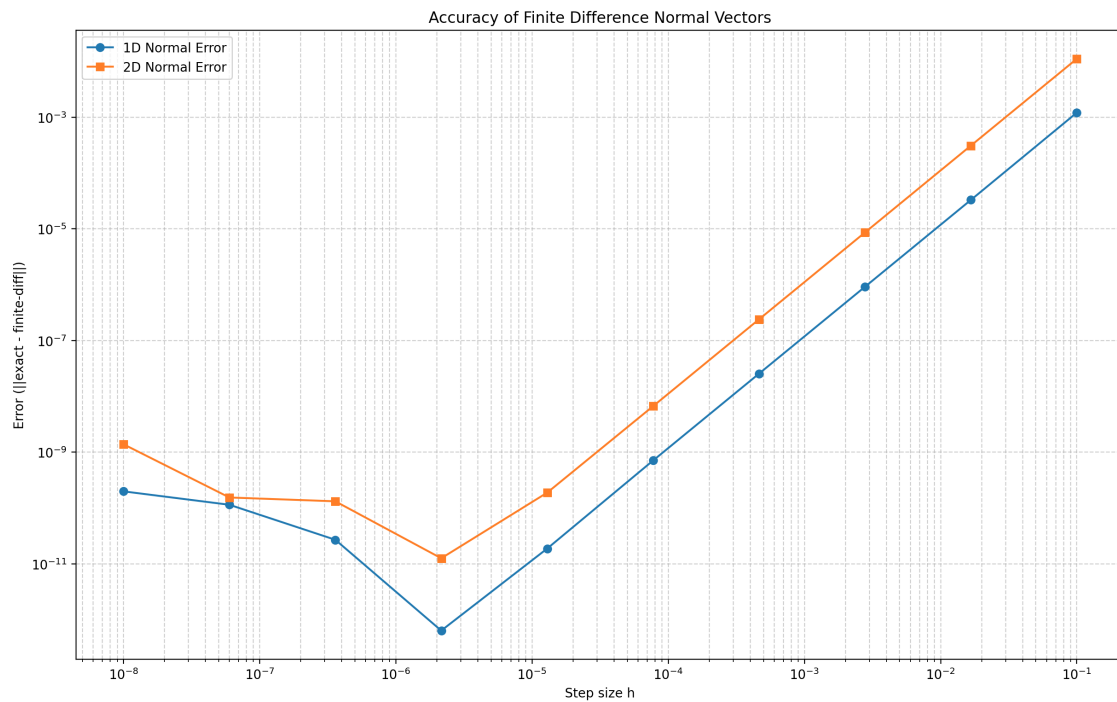


Figure 1: Error of finite difference normal vectors vs step size h (log-log scale).

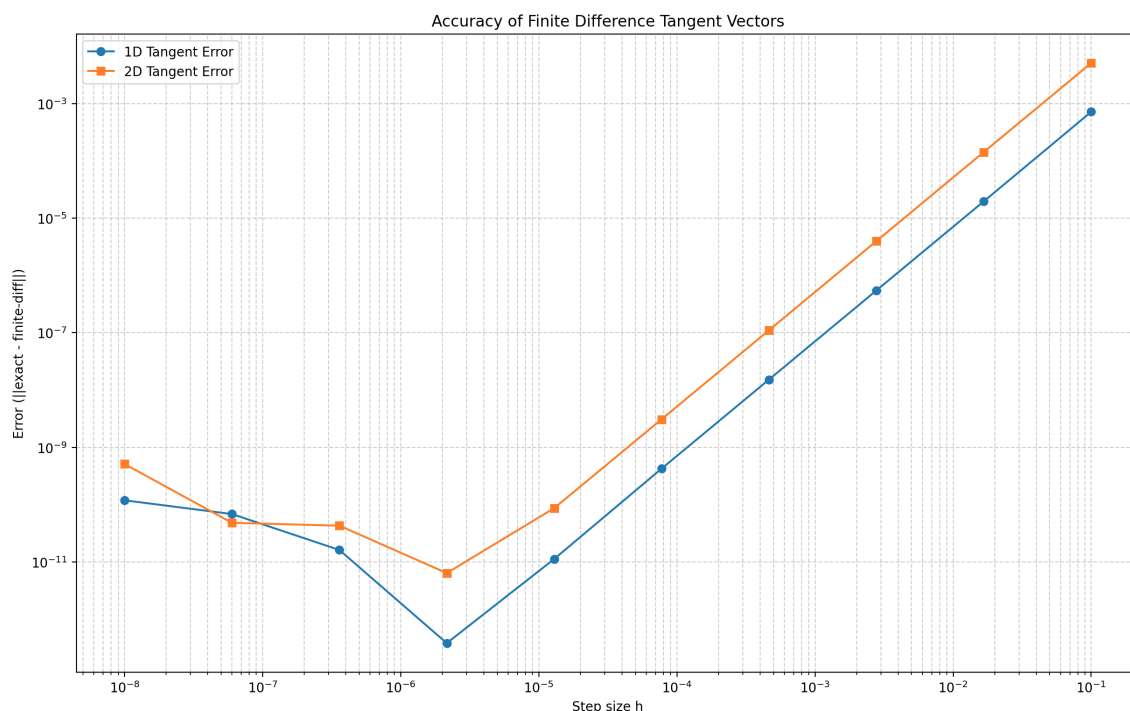


Figure 2: Error of finite difference tangent vectors vs step size h (log-log scale).

Observations

- Smaller h generally reduces error until round-off error dominates.
- 1D normal vectors show more stable convergence, while 2D normal vectors can oscillate slightly for very small h .
- Tangent vectors were also plotted: changing h changes their accuracy, but for sufficiently small h , the tangent vectors match the exact ones very well.
- Log-log plots help visualize the optimal h where finite difference is most accurate.
- Errors increase again for extremely small h due to numerical precision limits.

Conclusions

- For 1D functions and 2D surfaces, tangent lines and planes are easiest to get using the exact derivative.
- Using finite differences is a good way to approximate derivatives if we pick a sensible step size h .
- The normal vectors from finite differences get really close to the exact ones when h is not too small or too big.
- If h is too tiny, rounding errors mess things up. if it's too big, my approximation is less accurate.
- 2D gradients are trickier than 1D errors and sensitivity is higher in multi-dimensional cases.

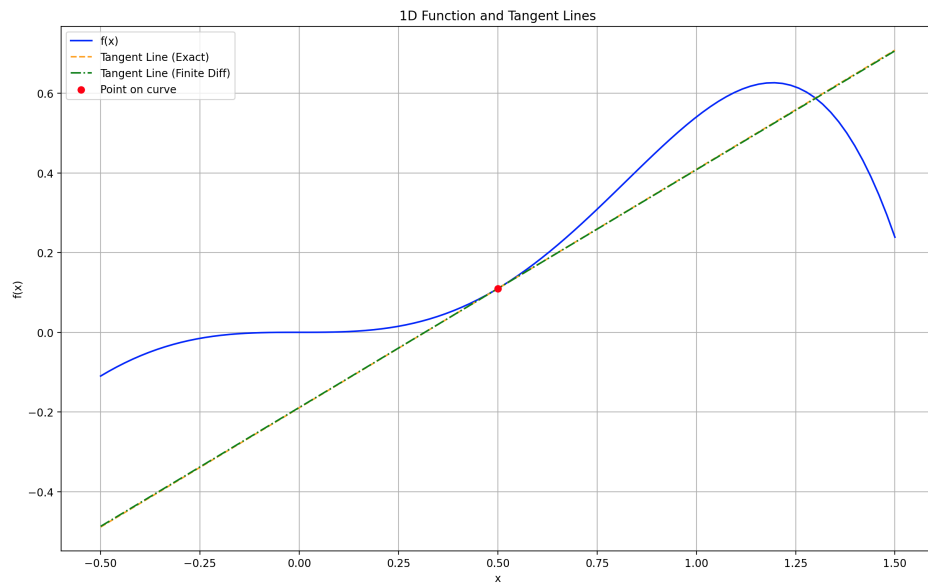


Figure 3: Tangent line for 2d(Exact and Finite)

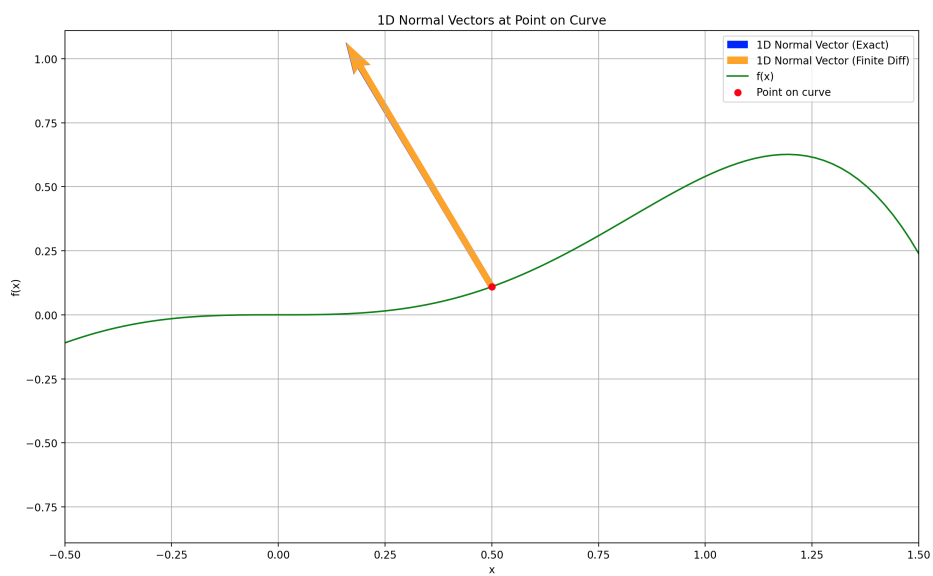


Figure 4: Normal Vector 2d(Exact and Finite)